

Jeffery Ye

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Software and AI engineer with experience designing, developing, and deploying full-stack systems (React, Django, SQL/NoSQL) and agentic AI solutions (LangGraph, PydanticAI).

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, R

AI/ML: Agentic AI, LangGraph, LangChain, PydanticAI, Large Language Models (LLMs)

Retrieval-Augmented Generation (RAG), LLM Evaluation, Natural Language Processing (NLP), PyTorch

Full-Stack: ReactJS, NodeJS, SQL (Postgres, MySQL), NoSQL (Neo4j, pgvector), Flask, FastAPI, Django

Platforms & DevOps: AWS, Spark, Databricks, GCP, Docker, Azure, Git, CI/CD, Linux/Unix, Hadoop

EDUCATION

BS Informatics (Focus: Software & AI) | University of Washington | Expected June 2026

- **Relevant Courses:** Data Structures/Algorithms, Adv. Databases, Web Dev., Adv. Data Science
- **GPA:** 3.55

PROFESSIONAL EXPERIENCE

Software/AI Research Assistant | Seattle Children's Hospital | April 2025 - Present

- Designed, developed, and deployed two multi-agent AI systems to automate complex research workflows, delivering tools that **reduced experimental design time by 95%**
- Took the initiative by **leading** brainstorming/design meetings, **interviewing** users, and **shadowing** stakeholders to iteratively improve upon the applications I developed.
- Engineered a scalable data pipeline integrating SQL, graph (Neo4j), and vector (pgvector) databases to synthesize insights from **1.2 TB** of unstructured genomic and proteomic data.
- Developed web interfaces with Django and Flask for researchers to interact with my applications.

Teaching Assistant (Informatics) | University of Washington | March 2025 - Present

- Delivered weekly hour-long lectures on Informatics concepts (e.g., AI, data science) to 50+ students, achieving a **4.9/5.0 rating** (Top 20% across all University of Washington instructors).
- Facilitated a **98% pass rate** through personalized support and one-on-one meetings, **mentoring 4 students** to be accepted into the competitive Informatics major.

TECHNICAL PROJECTS

AI Expense Categorizer

- Ideated and developed a fintech AI agent with PydanticAI to automate the secure classification of unstructured financial data, completing **~2 hours of manual work in 30 seconds**.
- Engineered a RAG architecture with IRS documentation for compliance and responsible AI usage.
- Used an evaluation framework with 500 transactions, achieving 98% classification accuracy.

Optum Claims Analysis | GRAIL (Biotech)

- Engineered SQL queries with JDBC to analyze 50k+ urinary cancer records in the Optum dataset.
- Synthesized findings into reports on treatment costs, utilizing Python (Pandas) for validation.

Musicalendar

- Developed a full-stack web application using ReactJS and a Flask back-end, deployed on AWS.
- Engineered a max-flow scheduling algorithm to optimize high-volume, 1-on-1 meeting bookings, replacing inefficient manual coordination via email chains and spreadsheets.

Educational Research STM | University of Utah

- Applied structural topic modeling to analyze **18,000+ educational policy research papers**, developing expertise in NLP pipelines, Spark, and statistical validation of topic model outputs.